MUBASHAR MEHMOOD

Portfolio: mehmoodm.github.io | GitHub: github.com/mehmoodm | LinkedIn: linkedin.com/in/mubashar-mehmood | mubasharmh145@gmail.com

EDUCATION

- **Master of Science in Computer Science** — **CGPA: 3.32/4.0**
  COMSATS University Islamabad, Pakistan — *2018 – 2021*
  Thesis: *A Classifier Model for Prostate Cancer Diagnosis Using CNNs and Transfer Learning with Multi-parametric MRI*
  Developed a deep learning model using CNNs and transfer learning to classify prostate cancer using multi-parametric MRI (mpMRI) data.
- **Bachelor of Science in Computer Science**
  COMSATS University Islamabad, Pakistan — *2013 – 2017*
  Final Year Project: *SnipBird: A Web Tool for Managing and Storing Research Content*
  Developed a web-based research management tool for capturing, organizing, and retrieving digital content using PHP, MySQL, JavaScript, and AJAX.

SCHOLARSHIPS & AWARDS

- **Punjab Educational Endowment Fund (PEEF) Scholarship** — *2018 – 2020*
  Merit-based scholarship awarded for the duration of the Master of Science in Computer Science program.

RESEARCH & DEVELOPMENT EXPERIENCE

Research Assistant — **COMSATS University Islamabad, Pakistan**

- Developed a deep learning framework for **prostate cancer classification** using multi-parametric MRI data.
- Researched **load balancing and resource allocation** algorithms in Cloud and Fog computing for distributed computing environments.
- Conducted **software refactoring** of legacy systems by identifying and addressing cross-cutting concerns to improve software modularity.

RESEARCH INTERESTS

Medical AI: Medical Image Analysis, Computer-Aided Diagnosis, Deep Learning for Medical Imaging, Image Classification
Computer Vision: Deep Learning, Transfer Learning, Image Representation Learning, Multimodal Learning
Emerging Interests: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Synthetic Data Generation

TECHNICAL SKILLS

- **Machine Learning & AI:** Machine Learning, Deep Learning, Computer Vision, Transfer Learning, Neural Networks
- **Medical AI:** Medical Image Analysis, Computer-Aided Diagnosis, Medical Image Classification, Multi-parametric MRI
- **Programming & Libraries:** Python, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, OpenCV
- **Data Analysis & Visualization:** Data Preprocessing, Exploratory Data Analysis, Matplotlib, Seaborn
- **Development Tools:** Jupyter Notebook, Google Colab, VS Code, Anaconda, Git/GitHub
- **Technical Writing:** LaTeX, Overleaf, Academic Writing, Technical Documentation

ACADEMIC PUBLICATIONS

Journal Articles

J.1. **Mehmood, M.**, Abbasi, S. H., Aurangzeb, K., Majeed, M. F., Anwar, M. S., & Alhussein, M. (2023). **A classifier model for prostate cancer diagnosis using CNNs and transfer learning with multi-parametric MRI**. *Frontiers in Oncology*, 13, 1225490. DOI

Conference Proceedings

C.1. **Mehmood, M.**, Javaid, N., Akram, J., et al. (2019). **Efficient resource distribution in cloud and fog computing**. In *Advances in Network-Based Information Systems (NBiS)*. Springer.

C.2. Ahmad, N., Javaid, N., **Mehmood, M.**, et al. (2018). **Fog-cloud based platform for utilization of resources using load balancing techniques**. In *International Conference on Network-Based Information Systems*.

C.3. Abbasi, S. H., Javaid, N., Ashraf, M. H., **Mehmood, M.**, et al. (2019). **Load stabilizing in fog computing environment using load balancing algorithm**. In *International Conference on Broadband and Wireless Computing (BWCCA)*. Springer.

SELECTED PROJECTS

- **Prostate Cancer Classification from Multi-parametric MRI:** Developed a deep learning model using CNNs and transfer learning for prostate cancer classification from multi-parametric MRI data. This work formed the basis of my MSc thesis and first-author journal publication.
- **Disease Prediction Models:** Developed and evaluated machine learning models including ANN, SVM, and Logistic Regression using publicly available datasets for diabetes, Parkinson's disease, and heart disease prediction.
- **Chronic Kidney Disease Predictor:** Developed an interactive web application using **Python, Streamlit, and Gradient Boosting** to predict chronic kidney disease risk from clinical patient data. Implemented the model and integrated it into a web-based interface for interactive prediction.
- **Chest X-Ray Classification:** Implemented a transfer learning approach using a fine-tuned DenseNet model for multi-class chest X-ray classification.

- **Skin Lesion Classification:** Developed a CNN-based model for classification of benign and malignant skin lesions.

CERTIFICATIONS TRAINING

- **AI for Medical Diagnosis** — DeepLearning.AI
- **Neural Networks and Deep Learning** — DeepLearning.AI
- **Building Deep Learning Models with TensorFlow** — IBM
- **Introduction to Computer Vision and Image Processing** — IBM
- **Machine Learning with Python** — IBM
- **Python for Data Science, AI & Development** — IBM
- **Generative AI for Healthcare** — Google Cloud
- **Generative AI and LLMs: Architecture and Data Preparation** — IBM
- **Large Language Models, Image Generation, and Responsible AI** — Google Cloud
- **Introduction to Deep Learning & Neural Networks with Keras** — IBM

ADDITIONAL INFORMATION

Languages	**English:** Professional Working Proficiency; **Urdu:** Native
Design Tools	Inkscape (Scientific Illustration & Data Visualization)

ACADEMIC REFERENCES

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